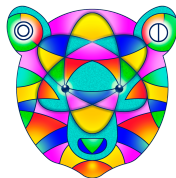
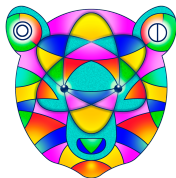


Higher moment theory and learnability of bosonic states

Joseph T. Josue

arXiv:2510.01610

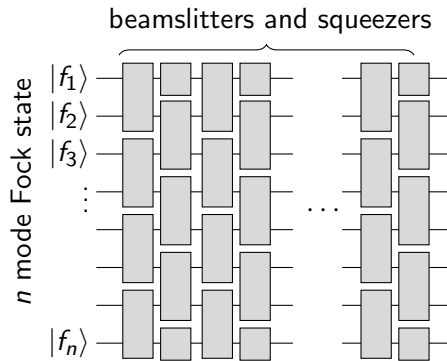
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- Learning (properties of) quantum states as efficiently as possible is a fundamental task in quantum information
- There is a lot of recent work on learning continuous variable (CV) states (eg. Mele, Mele, et al. 2024; Zhao, Liao, et al. 2024; Bittel, Mele, et al. 2025; Fanizza, Iyer, et al. 2025)
- Most *sample and time efficient* learning algorithms work for **Gaussian** or **near-Gaussian** states
- Learning a generic n -mode CV state requires $\sim (\text{energy})^n$ samples (Mele, Mele, et al. 2024)

Our goal

- Our goal is to find interesting, **highly non-Gaussian** classes of CV states with *sample and time efficient* learning algorithms
- Motivated by an open problem proposed in Aaronson and Grewal 2023, we start with **BosonSampling** states



Background

- A bosonic system of n modes is described by the creation and annihilation operators $a_1^\dagger, \dots, a_n^\dagger, a_1, \dots, a_n$
- Equivalently, by position and momentum operators $\mathbf{r} = (x_1, \dots, x_n, p_1, \dots, p_n)$
- A unitary transformation is called *Gaussian* if it acts affinely on $\mathbf{r}$

$$\mathcal{U}_{S,\mathbf{d}}^\dagger r_i \mathcal{U}_{S,\mathbf{d}} = d_i + \sum_j S_{ij} r_j$$

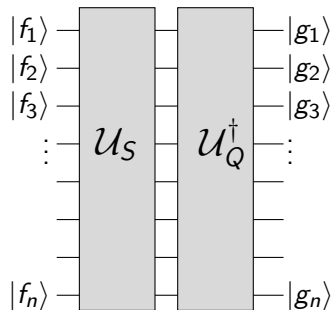
- **The group of Gaussian unitaries is $\mathrm{Sp}(2n, \mathbb{R}) \ltimes \mathbb{R}^{2n}$**

Problem statement (Aaronson and Grewal 2023)

- Let $|\mathbf{f}\rangle$ denote an unknown Fock state on n modes,
 $\mathbf{f} = (f_1, \dots, f_n) \in \mathbb{N}_0^n$
- Let $\mathcal{U}_S$ denote a Gaussian unitary specified by an unknown matrix $S \in \text{Sp}(2n, \mathbb{R})$
- Let $|\psi\rangle = \mathcal{U}_S |\mathbf{f}\rangle$

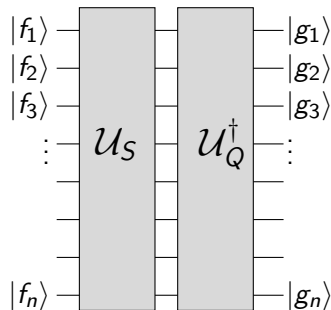
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- Let $|\psi\rangle = \mathcal{U}_S |\mathbf{f}\rangle$
- Given N copies of ψ , we want to learn ψ to ε precision
- Find a $\mathbf{g}$ and Q such that $|\langle \mathbf{g} | \mathcal{U}_Q^\dagger \mathcal{U}_S | \mathbf{f} \rangle| \geq 1 - \varepsilon$



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- | | |
|---|---|
| • How large must N be? | <i>Want it to be $\leq \text{poly}(\text{energy})$</i> |
| • What is the measurement scheme? | <i>Want it to be practically accessible</i> |
| • How hard is the classical postprocessing? | <i>Want it to be $\leq \text{poly}(n)$</i> |

Analogous fermionic problem

- The analogous fermionic problem was solved in Aaronson and Grewal 2023
- Crucial difference: fermionic Fock states are Gaussian, bosonic Fock states are highly non-Gaussian

First main result

- We will solve this problem with *moments* for $|\psi\rangle = \mathcal{U}_S |\mathbf{f}\rangle$

$$\Lambda_{i_1, \dots, i_t; j_1, \dots, j_t}^{(t)} = \langle \psi | r_{i_1} \dots r_{i_t} r_{j_1} \dots r_{j_t} | \psi \rangle$$

- Suppose we know $\Lambda^{(1)'}, \Lambda^{(2)'}$ such that $\|\Lambda^{(t)'} - \Lambda^{(t)}\| \leq \varepsilon_t$

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Theorem

We find an **efficient** algorithm that finds a Q and $\mathbf{g}$ such that

$$|\langle \mathbf{f} | \mathcal{U}_S^\dagger \mathcal{U}_Q | \mathbf{g} \rangle| \geq 1 - \mathcal{O}\left(\varepsilon_1 n^2 \|\mathbf{f}\|_\infty^3 \|S^T S\|^{3/2} + \varepsilon_2 n^2 \|\mathbf{f}\|_\infty \|S^T S\|^3\right)$$

This can therefore be combined with *any* strategy to measure moments to yield a **sample and time efficient learning algorithm**

Measuring moments

Theorem

Using a **median of means** estimator, $\Lambda^{(2t)}$ for $\mathcal{U}_S |\mathbf{f}\rangle$ can be estimated to norm precision ε with probability at least $1 - \delta$ using $N \sim \mathcal{O}\left(\frac{\log(n/\delta)}{\varepsilon^2} (n\|f\|_\infty \|S^T S\|)^{2t}\right)$ **heterodyne** samples.

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Corollary (Full solution to Aaronson and Grewal 2023)

N heterodyne samples from the state $\mathcal{U}_S |\mathbf{f}\rangle$ suffice via our efficient algorithm to learn a $Q, \mathbf{g}$ satisfying $|\langle \mathbf{f} | \mathcal{U}_S^\dagger \mathcal{U}_Q | \mathbf{g} \rangle| \geq 1 - \gamma$ with probability at least $1 - \delta$, where

$$N \sim \mathcal{O}\left(\frac{\log(n/\delta)}{\gamma^2} n^8 \|\mathbf{f}\|_\infty^{10} \|S^T S\|^{10}\right)$$

Roughly, $n \|\mathbf{f}\|_\infty \|S^T S\| \sim \text{energy}$

Overview

1 Introduction and main results

2 Sample efficiency

3 Time efficiency

4 Outlook

Definition

A G_t state is a thermal (or ground) state of a (non-degenerate) Hamiltonian that is degree t in the quadrature operators

- A G_t state is **fully specified** by its first t moments

Pure G_t states

- Suppose ψ is a G_t state; it is the ground state of $H = \sum_i \alpha_i \hat{M}_i$ where $\hat{M}_i$ are degree $\leq t$ operators

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$$\begin{aligned}\langle \psi | H | \psi \rangle &= \min_{\phi} \sum_i \alpha_i \langle \phi | \hat{M}_i | \phi \rangle \\ &= \min_{m_1, m_2, \dots} \sum_i \alpha_i m_i \quad \text{st } \exists \phi, \langle \phi | \hat{M}_i | \phi \rangle = m_i\end{aligned}$$

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- Because H is non-degenerate, any state with the matching moments is ψ
- If H is *gapped*, then **inverse poly moment precision corresponds to inverse poly state fidelity** (Yu and Wei 2023)

Relevance to the learning algorithm

- A Fock state $|\mathbf{f}\rangle$ is a G_4 state; it is the ground state of $H = \sum_i (\hat{a}_i^\dagger \hat{a}_i - f_i)^2$
- The set of G_t states is closed under the action of Gaussian unitaries (because they act affinely)

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- Therefore, $\mathcal{U}_S |\mathbf{f}\rangle$ is the ground state of a gapped degree 4 Hamiltonian
- Knowing the **first four moments to inverse poly precision** suffices to learn the state
- But this does *not* guarantee a time-efficient learning algorithm

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Sketch of our algorithm

- For simplicity, we will restrict our attention here to
passive Gaussian unitaries (specified by $W \in \mathcal{U}(n)$)
the Fock state $|1, \dots, 1\rangle \equiv |1^n\rangle$
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the error free case (assume moments are known *perfectly*)
- We sketch the learning algorithm with these simplifications

Passive moment transformations

- Recall $W \in U(n)$ acts as

$$\mathcal{U}_W^\dagger a_i \mathcal{U}_W = \sum_j W_{ij} a_j$$

- How do moments **transform**?

$$\sigma_{i_1, \dots, i_t; j_1, \dots, j_t}^{(t)} = \langle 1^n | \mathcal{U}_W^\dagger a_{i_1} \dots a_{i_t} a_{j_1}^\dagger \dots a_{j_t}^\dagger \mathcal{U}_W | 1^n \rangle$$

$$(\sigma_0^{(t)})_{i_1, \dots, i_t; j_1, \dots, j_t} = \langle 1^n | a_{i_1} \dots a_{i_t} a_{j_1}^\dagger \dots a_{j_t}^\dagger | 1^n \rangle$$

$$\sigma^{(t)} = W^{\otimes t} \sigma_0^{(t)} W^{\dagger \otimes t}$$

- $\sigma^{(t)}$ is a submatrix of a unitary transformation of the $\Lambda^{(t)}$ moments

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- So now we use **fourth moments**, $\sigma^{(2)}$

Learning algorithm

Given the fourth moments $\sigma^{(2)} = W^{\otimes 2} \sigma_0^{(2)} W^{\dagger \otimes 2}$ for the state $\mathcal{U}_W |1^n\rangle$, determine W

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- Because $\sigma_0^{(2)}$ is a matrix on $(\mathbb{C}^n)^{\otimes 2}$, we can represent it in bra-ket notation using the standard basis $|1\rangle, \dots, |n\rangle$ of $\mathbb{C}^n$

$$\sigma_0^{(2)} = 4(\mathbb{I} + U_{\text{SWAP}}) - 2 \sum_{i=1}^n |i, i\rangle \langle i, i|,$$

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- Denote the i^{th} column vector of W by $|w_i\rangle$. Given access to $\sigma^{(2)}$, we can thus form the matrix

$$\begin{aligned} A &= \frac{1}{2} \left(4(\mathbb{I} + U_{\text{SWAP}}) - \sigma^{(2)} \right) \\ &= \sum_{i=1}^n (|w_i\rangle \otimes |w_i\rangle)(\langle w_i| \otimes \langle w_i|), \end{aligned}$$

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Given access to a matrix $A = \sum_{i=1}^n (|w_i\rangle \otimes |w_i\rangle)(\langle w_i| \otimes \langle w_i|)$, learn $|w_i\rangle$ (up to phases and permutations)

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- Unitarily diagonalize A and take the $+1$ eigenvalues — we learn the eigenspace $\text{span}\{|w_i\rangle \otimes |w_i\rangle \mid i = 1, \dots, n\}$
- A numerical diagonalization routine will return the vectors

$$|\tilde{w}_i\rangle = \sum_{j=1}^n U_{ij} |w_j\rangle \otimes |w_j\rangle = \sum_{j=1}^n |U_{ij}| e^{i\phi_{ij}} |w_j\rangle \otimes |w_j\rangle$$

- The unitary U is totally arbitrary and unpredictable

Learning algorithm

Given access to the vectors $|\tilde{w}_i\rangle = \sum_{j=1}^n |U_{ij}| e^{i\phi_{ij}} |w_j\rangle \otimes |w_j\rangle$ with U, ϕ unknown and arbitrary, learn $|w_j\rangle$ (up to phases and permutations)

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- Take the intersection of the Schmidt decompositions of all $|\tilde{w}_i\rangle$ to learn all the $|w_i\rangle$ up to phases

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- Take the intersection of the Schmidt decompositions of all $|\tilde{w}_i\rangle$ to learn all the $|w_i\rangle$ up to phases
- We define V to be the unitary matrix whose columns are precisely these learned vectors
- By construction, V is equal to W up to a permutation of its columns and global phases applied to the columns

- In other words, we have learned the matrix $V = W\Phi P$, where Φ is some arbitrary diagonal unitary matrix, and P is some arbitrary permutation matrix
- It follows that the state $\mathcal{U}_V |1^n\rangle$ is the same as $\mathcal{U}_W |1^n\rangle$ up to an irrelevant global phase, thereby completing the learning algorithm

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- To prove the main result, we need to track how the error in the moments propagates through the algorithm to an error in the fidelity

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- 4 Outlook**

Summary

- By going beyond second moments, we found a **sample and time efficient** algorithm to learn *BosonSampling* states
- Solves open problem left by Aaronson and Grewal 2023

Summary

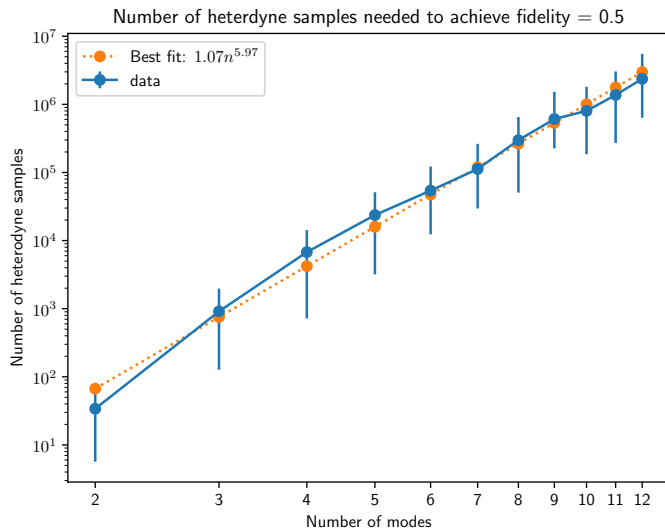
- By going beyond second moments, we found a **sample and time efficient** algorithm to learn *BosonSampling* states
- Solves open problem left by Aaronson and Grewal 2023
- We construct an **infinite set of Gaussian invariants for generic bosonic states** using classical invariant theory of the action of symplectic group on moments, relevant for state conversion (Chabaud, Markham, et al. 2020; Hahn, Ferrini, et al. 2024)

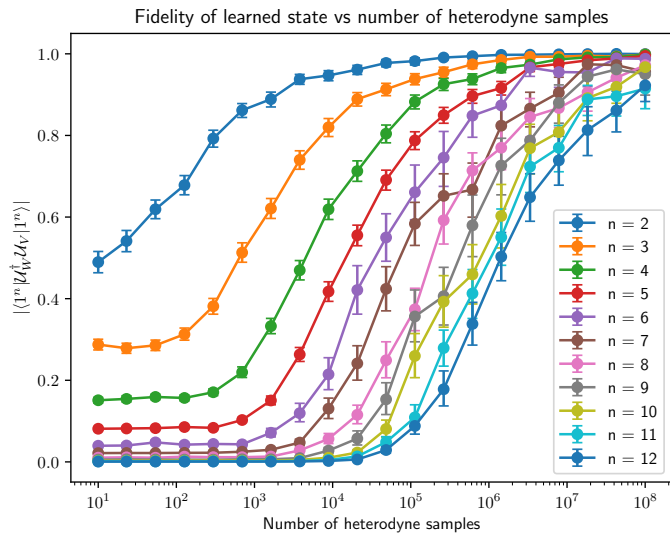
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- Lots of possible extensions!

- Our algorithm generalizes to a slightly larger class of states, and fails interestingly in other places
- Certain classes of states *hide information* from lower moments
- Invariants and generalizations of Wick's theorem

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- Certain classes of states *hide information* from lower moments
- Invariants and generalizations of Wick's theorem
- Can we make our algorithm have more favorable scaling with energy?
 - Our algorithm requires $N \sim \text{poly}(n, \|\mathbf{f}\|_\infty, \|S^T S\|)$ samples
 - Applying (Bittel, Mele, et al. 2025), we get $N \sim \text{poly}(n, \|\mathbf{f}\|_\infty, \log\|S^T S\|)$

Additional slides





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